

Assignment 1 (Weeks 1 and 2)

Instructions

- This assignment covers material from Weeks 1 and 2.
- Submit a single PDF file containing all three parts of the assignment.
- For Part III, draw your ER diagram in draw.io, available free at www.diagrams.net. Once done, export it as a PNG (File → Export As → PNG, with Fit Page enabled), insert it into your Word/Latex document along with your answers and export the whole document as a PDF.
- All written answers should be your own. Do not copy definitions directly from resources.

PART I : Short Answer Questions

Q1. What is the difference between a Database Management System (DBMS) and a traditional file system? Provide one concrete scenario in which a DBMS is clearly the better choice and explain your reasoning.

—A special software program that helps users to create, manage, and organize Databases.

A traditional file system stores data in separate files managed by the operating system.

A university stores student records, course, grades, and faculty information. A DBMS is a better choice because multiple users can access the data simultaneously, relationships between entities can be maintained, and data redundancy is reduced

Q2. In an ER diagram, what distinguishes a weak entity from a strong entity? How is the primary key of a weak entity formed and what is the identifying relationship associated with it called?

—A strong entity has its own primary key and can be uniquely identified independently.

A weak entity cannot be uniquely identified using its own attributes and depends on another entity.

The primary key of a weak entity is formed by primary key of the owner entity & partial key (discriminator) of the weak entity.

Q3. A university database stores the phone numbers of students, where a student may have multiple phone numbers. Should the phone number be modeled as a simple attribute, a composite attribute or a multivalued attribute? Justify your answer.

—Phone number should be modeled as a multivalued attribute.

A student may have more than one phone number. Since multiple values can exist for the same student, the attribute is multivalued.

Q4. Explain the difference between a candidate key and a super key. Consider the relation: Student(roll no, email, name, branch) List all candidate keys and provide at least two super keys that are not candidate keys.

–Candidate Key: A minimal set of attributes that uniquely identifies a tuple.

Super Key: Any set of attributes that uniquely identifies a tuple. It may contain extra attributes.

- Candidate Keys:

Roll_no

email

- Super Keys (not candidate keys):

(roll_no, name)

(email, branch)

(roll_no, email)

(roll_no, email, name)

Q5. Suppose a row is inserted into an Enrollment table with a student id value that does not exist in the Student table. • Name the constraint that is violated. • Explain what it protects against.

- Name the constraint that is violated.

Constraint Violated is Foreign Key Constraint .

It prevents records from referencing non-existent entities and maintains consistency between related tables.

- Explain what it protects against.

It prevents records from referencing non-existent entities and maintains consistency between related tables.

- State what should happen to the insert operation

The insertion should be rejected by the DBMS and an error should be generated because the referenced student does not exist in the Student table.

Q6. When mapping a many-to-many (M:N) relationship from an ER diagram to a relational schema:

- What additional structure must be created?

A junction table (associative table).

- What does its primary key consist of?

The combination of the primary keys of the participating entities.

- Why can the relationship not simply be represented as a column in one of the participating entity tables?

A single column cannot represent multiple associations efficiently. Doing so would create redundancy and violate normalization principles. The junction table properly represents all possible combinations between the entities.

Part II : Long Answer Question

An online food delivery platform needs to store information about customers, restaurants, menu items and orders.

- A customer can place multiple orders.
- Each order is associated with exactly one restaurant.
- An order may contain multiple menu items, each with a specified quantity.
- A restaurant can offer multiple menu items.

- **Part A – Entity and Relationship Identification**

Entity and Relationship Identification Identify all entities and their key attributes. For each relationship:

- Specify its cardinality (1:1, 1:N or M:N).
- Justify your choice in one sentence.

Customer == CustomerID (PK),Name,Email,PhoneNumber.

Cardinality: 1:N – One customer can place many orders, but each order is placed by exactly one customer.

Restaurant == RestaurantID (PK),Name,Address,ContactNumber.

Cardinality: 1:N – One restaurant can receive many orders, but each order belongs to exactly one restaurant.

Order == OrderID (PK), OrderDate, TotalAmount.

Cardinality: 1:N – A restaurant can offer many menu items, and each menu item belongs to one restaurant.

MenuItem == MenuItemID (PK), ItemName, Price.

Cardinality: M:N – An order may contain multiple menu items, and the same menu item may appear in many different orders.

- **PART B – Attribute Classification**

Identify all entities and their key attributes. For each relationship:

- Specify its cardinality (1:1, 1:N or M:N).
- Justify your choice in one sentence.

Multivalued Attribute == Customer PhoneNumber – A customer may have multiple phone numbers; therefore multiple values can be stored for the same customer.

Composite Attribute == Restaurant Address – Since it consists of multiple meaningful subparts, it is a composite attribute. Address can be divided into: House Number, Street, City, State, PIN Code.

- **Part C – Relational Schema Design**

The relationship between Order and MenuItem is M:N.

Write the complete relational schema for the junction table created to represent this relationship, including:

- Primary key.
- Foreign keys.
- Any additional attributes required.

Use the notation:

TableName(col1, col2, col3)

OrderMenuItem(OrderID, MenuItemID, Quantity)

PK: (OrderID, MenuItemID)

FK: OrderID → Order(OrderID)

MenuItemID → MenuItem(MenuItemID)

Part III : ER Diagram Design.

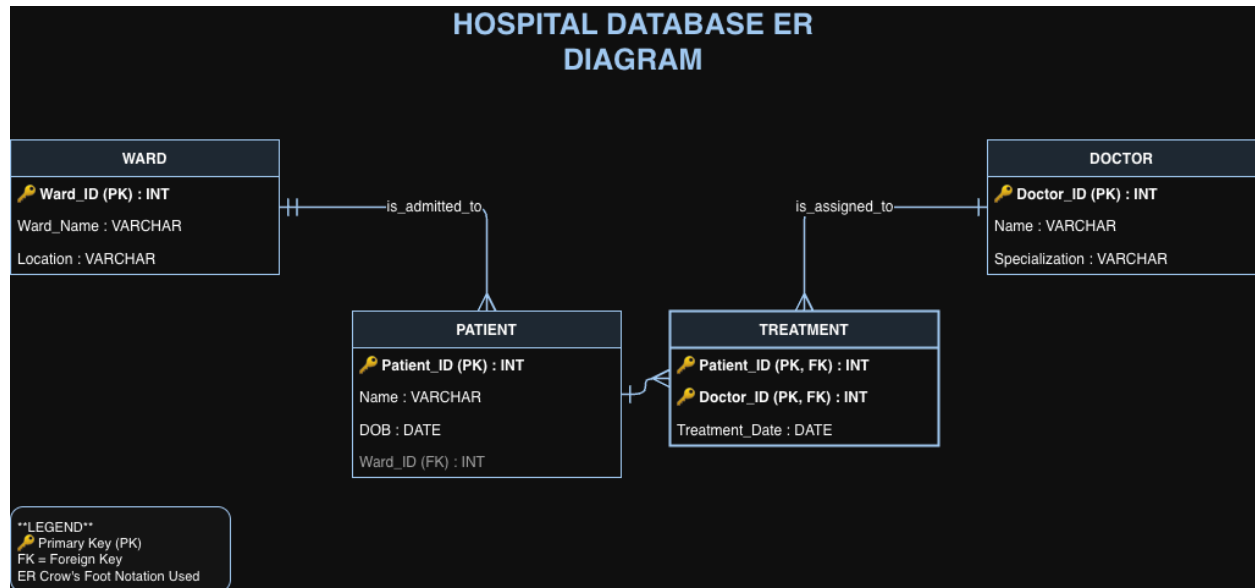
Question 1: Hospital ER Diagram

Draw an ER diagram using draw.io based on the following hospital scenario:

- The database must track Patients, Doctors and Wards.
- A ward can house many patients, but a patient is admitted to only one ward at a time.
- A patient can be treated by multiple doctors during their stay.
- A doctor can treat multiple patients.

Your ER diagram must include:

- Appropriate attributes for each entity.
- A clearly marked primary key for every entity.
- All relationships with the correct cardinality (1:1, 1:N, or M:N).
- Participation constraints wherever applicable.



Question 2: Design Justification

Write a justification (150–200 words) addressing all three of the following points:

- The Doctor–Patient relationship is M:N in this scenario. Explain why this is the case, identify the associative entity it creates and state what its composite primary key would consist of.
- For at least two relationships in your diagram, state whether each participating entity has total or partial participation and explain your reasoning. For example, must every patient be admitted to a ward or could a patient exist in the system without being assigned to one?
- One attribute design decision you made explain why you modelled a particular attribute the way you did (e.g. why an address was made composite or why a certain attribute is multivalued rather than single-valued).

The Doctor–Patient relationship is many-to-many (M:N) because a doctor can treat multiple patients, and a patient may be treated by multiple doctors during their stay in the hospital. For example, a patient may consult a surgeon, a physician, and a specialist, while each doctor is responsible for several patients. This M:N relationship creates an associative entity called Treatment. The composite primary key of this entity consists of DoctorID and PatientID, which together uniquely identify each doctor–patient association.

In the Patient–Ward relationship, the Patient entity has total participation because every admitted patient must be assigned to a ward. The Ward entity has partial participation since a ward can exist even when no patients are currently admitted. In the Doctor–Patient relationship, both Doctor and Patient have partial participation because a doctor may temporarily have no assigned patients, and a patient record may exist before treatment begins.

One attribute design decision was modeling Address as a composite attribute. Address can be divided into meaningful components such as house number, street, city, state, and postal code. This improves data organization, allows more efficient searching and reporting, and makes updates easier when only one part of the address changes.